Case Problem: Hanover Inc.

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May 5, 2022

Executive summary

The purpose of this case analysis is to investigate whether Hanover Inc. needs to build a new factory to fulfill the predicted demand for the next five years and which location can provide the optimal solution to create the most profit. With the available information about factory capacity, operation cost, production cost, shipping cost and duty cost, it is recommended to build a new factory due to the fact that the current factory will not be able to fulfill the predicted demand. The analysis report uses linear programming to look for the optimal solution which can minimize the total cost for the production. While maintaining the operation of existing factories, one of the factories is identified as the best location if Hanover, Inc. decided to build a new factory in the future. While the possible expense regarding building a plant at a new location is unclear. It is recommended to conduct further investigation in the future.

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Introduction

Hanover Inc. is in the manufacturing industry that produces fiber optics products. With the increasing global demand, Hanover Inc. predicts the demand from their customers will increase in the next five years for their products A, B and C. Currently, Hanover Inc. has two factories that are capable of producing product A, B and C in Austin and Paris. They provide the products to customers in Malaysia, China, France, Brazil, US Northeast, US Southeast, US Midwest, and US West. With the existing factories of Austin and Paris with the capacity of 600,000 units each, Hanover Inc. requested an analysis to determine whether they should build a new factory to fulfil the upcoming demand.

The possible factory location includes Charleston SC, Mobile AL in the United States and other overseas locations in Australia, India, Malaysia, South Africa, and Spain. Hanover Inc. provides the following information for the analysis: Fixed cost with different capacity for each factory; production cost for each product from each factory; shipping cost from factory to all the customers; selling price for each product for all the customers; duty cost for each county and the forecast demand in the next five years per year.

Case Background Analysis

The purpose of this case study is to identify if Hanover, Inc needs to establish a new factory to supply for its customers based on the forecast of demand in the next five years. If the current two existing factories do not have the capacity or cost too much to increase the capacity for the production, then a new factory location needs to be identified.

Product Cost and Demand

Hanover, Inc produces 3 different products with different production costs for each. The following table is the production price for existed factories Austin and Paris as well as candidate factories(figure.1):

Figure.1
Production Cost for each factory

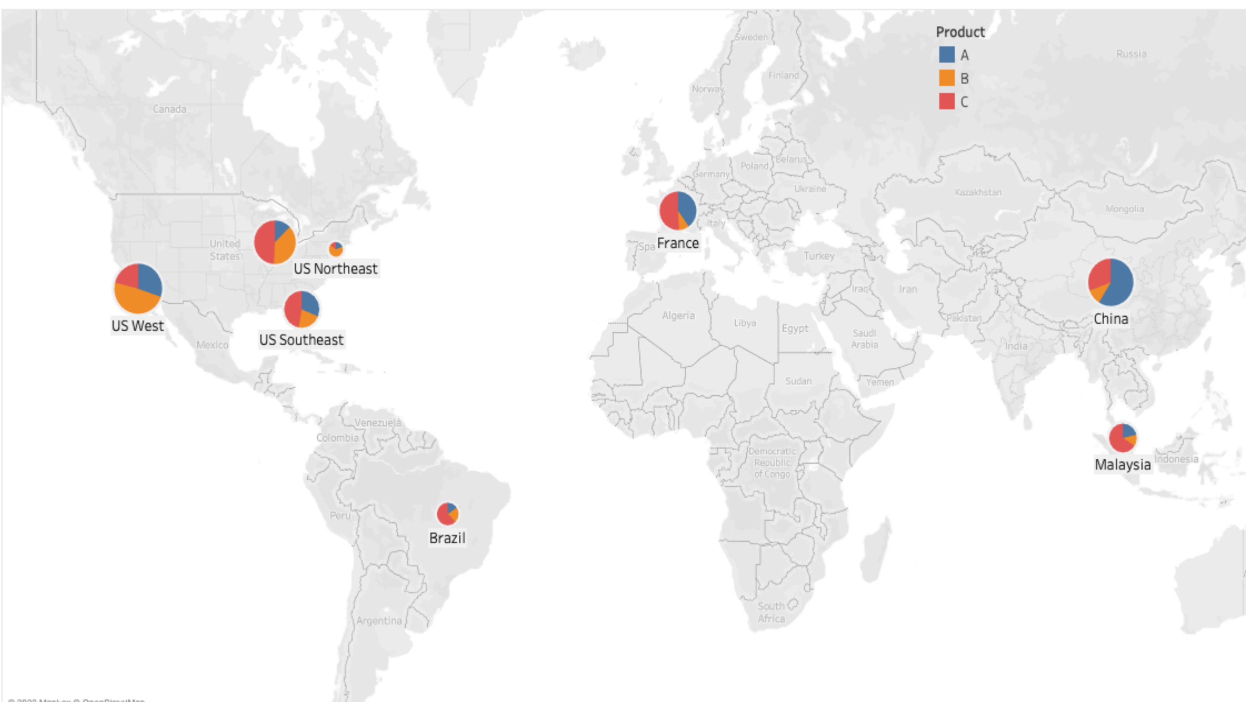
	A	B	C
Austin	$1.50	$1.50	$1.56
Paris	$1.56	$0.91	$0.89
Australia	$0.75	$1.74	$1.70
Charleston	$1.44	$0.90	$0.88
India	$0.75	$0.96	$0.95
Malaysia	$0.80	$1.25	$1.65
Mobile	$1.25	$1.57	$1.55
S Africa	$0.76	$0.92	$0.90
Spain	$1.56	$1.88	$1.86

It appears that although Austin and Paris factories are currently serving the demand from the customers, their production cost is not the cheapest. For product A, Paris is the most expensive factory to produce product A, followed by Austin and Spain. The factories that can spend the least money to produce product A are Australia and India. As for product B, the most expensive factory to produce it is Span, followed by Austin. The factory that can spend the least cost to produce product B is Charleston. Finally, as for product C, Spain appears to be the factory that needs to spend the most money to produce it, followed by Australia. The cheapest option to produce product C will be Charleston. Additionally, the cost to produce product C from Spain and Australia is significantly more expensive than the cheapest option. For Spain, it will cost more than twice the price from Charleston.

The following chart demonstrates the demand from each customer in a geographical map with pie charts to represent the approximate needs for each product (figure.2). It appears that the United States area potentially needs a higher quantity of product A compared to other customers. At the same time, China and France appear to have higher demand in product C. The total demand from Malaysia and Brazil seems smaller than the other customers.

Figure.2

Map for Customer Demand

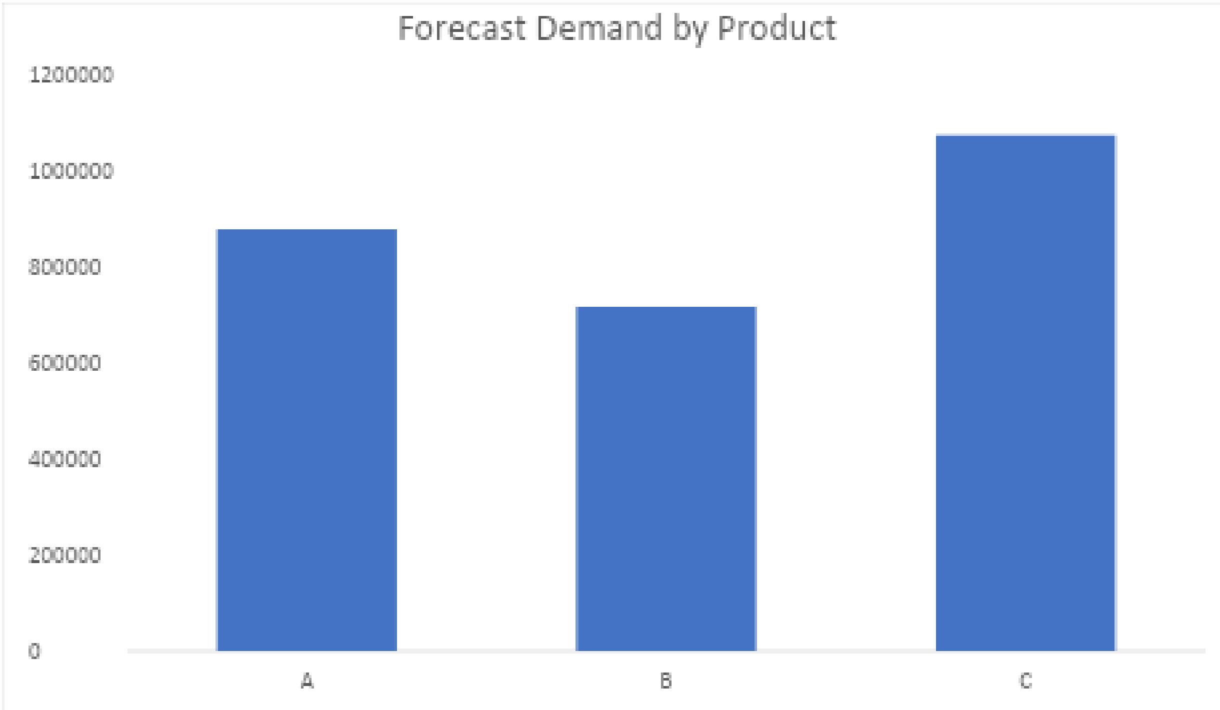


Furthermore, the total demand from product A, B, C is shown in the following chart (Figure 3).

Product C appears to have the highest demand of all and product A the second.

Figure.3

Forecast Demand by Product



Factory Cost and Capacity

The Hanover. Inc. currently have Austin and Paris factories to produce A, B, C products for the customer. While Hanover, Inc. is expecting the potential needs for the future demand, it has a list of candidate factories that can potentially be the optimal solution: Australia, Charleston, Mobile, India, South Africa, Malaysia, and Spain. The factories have different costs for maintaining the operation which includes capital costs, insurance, management, etc. based on the capacity of 600,000 or 1,200,000. The following two graphs show the fixed cost for each factory, it appears that Mobile cost the most with the capacity of 600,000 while the existing factories Austin and Paris are the cheapest (Figure 4 & Figure 5). As for the capacity of 1,200,000, the most expensive option will be the Charleston location while Paris is the cheapest.

Figure 4.

Operation Cost for Each for the capacity of 600,000

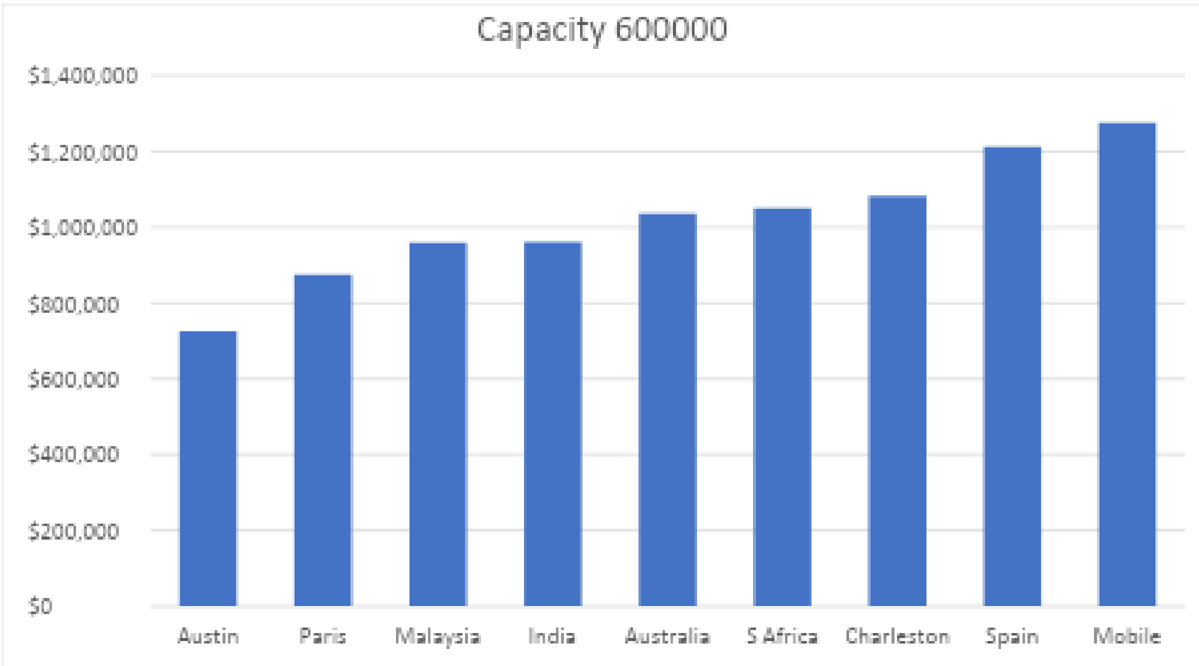
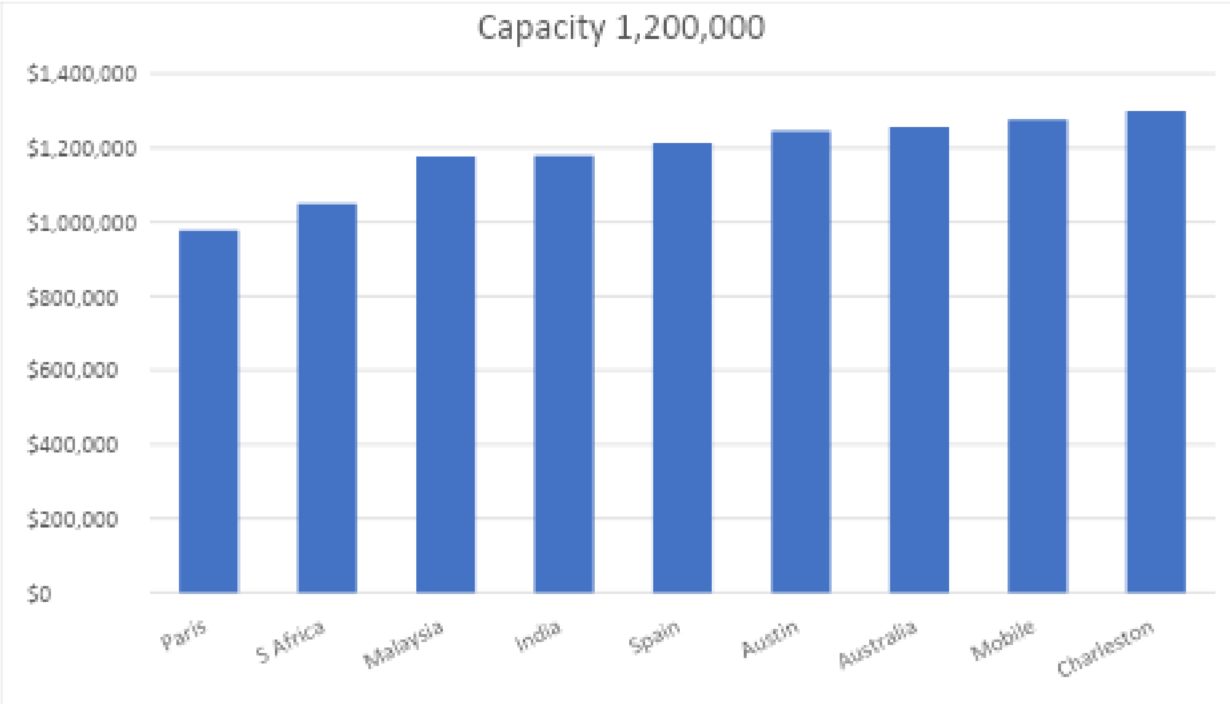


Figure 5.

Operation Cost for Each for the capacity of 1,200,000



Only investigating the rank for the fixed cost for operating the factory may not be sufficient to make the best option when deciding which factory should have the capacity of 600,000 and which factory should have the capacity of 1,200,000. Therefore, the analysis will use the Benefit-Cost Ratio (BCR) to compare the expenses associated with expansion in each factory. BCR is a measurement to evaluate how much benefit will be obtained with the expense. The calculation of BCR is using the present value of benefit divided by the present value of cost. Therefore, the BCR for Hanover, Inc.'s factory: Factory Capacity/ Fixed Cost of the factory= BCR. The following table shows the BCR of each factory with different capacity:

BCR	Austin	Paris	Australia	Charleston	India	Malaysia	Mobile	S Africa	Spain
600000	0.83	0.69	0.58	0.55	0.63	0.63	0.57	0.75	0.60
1200000	0.96	1.27	0.78	0.97	1.10	1.00	0.92	1.22	0.87

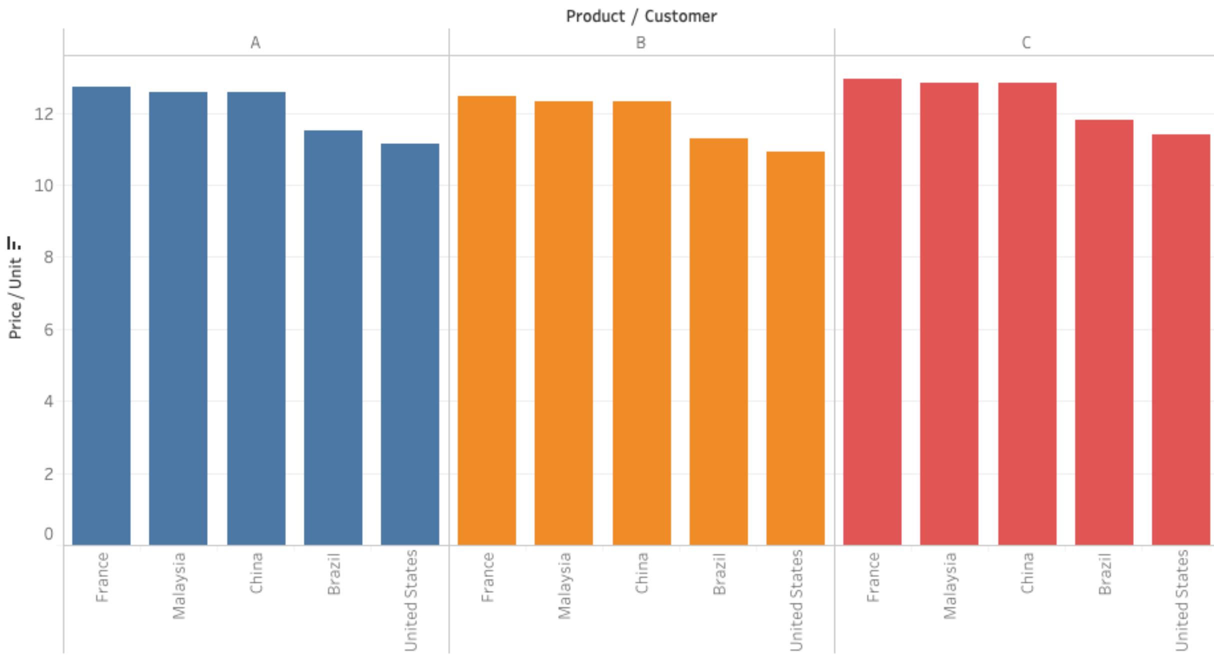
According to the table above, Austin and Paris appear to have better BCR with the capacity of 1200000 than 600000. Therefore, the choice of expanding the capacity of existing factories should be considered during the process of deciding the best candidate for the new factory. On the other hand, other candidate factories all have better BCR with 1200000, however, India and South Africa stand out from other factories with their BCR more than which indicates that these two factories may earn more benefit with the associated cost.

Customer Cost and Demand

The Hanover, Inc. currently has eight customers: Malaysia, China, France, Brazil, US northeast, US Midwest, US southeast and US west. According to the forecast they all have different demand for different products. Hanover, Inc sells their three different products with different prices in each country. However, Hanover, Inc. needs to pay for the duty cost for each country which is based on the total sales. The following table shows the selling price for each product in each country (figure 6). It appears that the selling price for all the products in the United States is the lowest amount. The selling price for all the products in Paris is the highest. Additionally, product C also has the highest selling price compared to other two products.

Figure 6

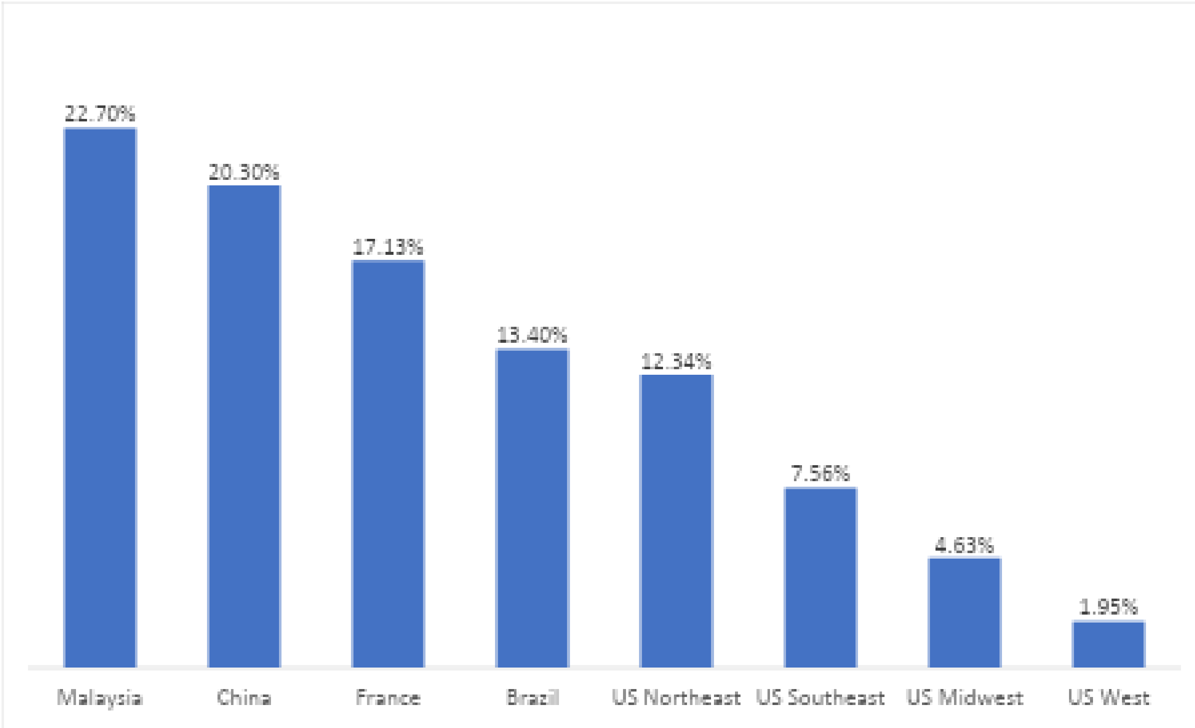
Selling Price for different product in each country



According to the forecast for the demand in the next five years, the demand from each country is different between products. The following chart is the percentage of total demand of all the products from each country (Figure7&8). The customer has the highest demand in Malaysia, followed by China, France, Brazil, and each region in the United States. However, if the demand from each region from the United States were combined, the total demand from the United States is the highest of all the customers. Therefore, while Hanover Inc. is evaluating the decision of opening a new factory to support the future demand, the needs from Malaysia or the United States may need to be considered with priority due to their higher demand.

Figure 7.

Demand Percentage for Each Customers



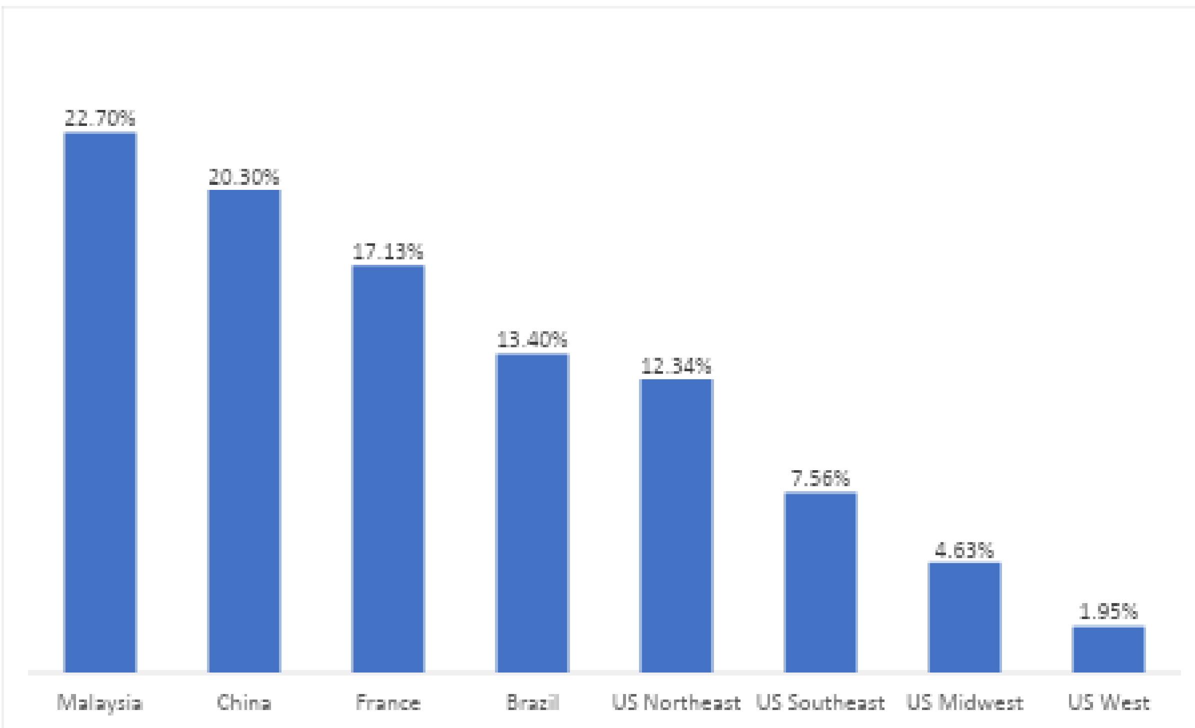
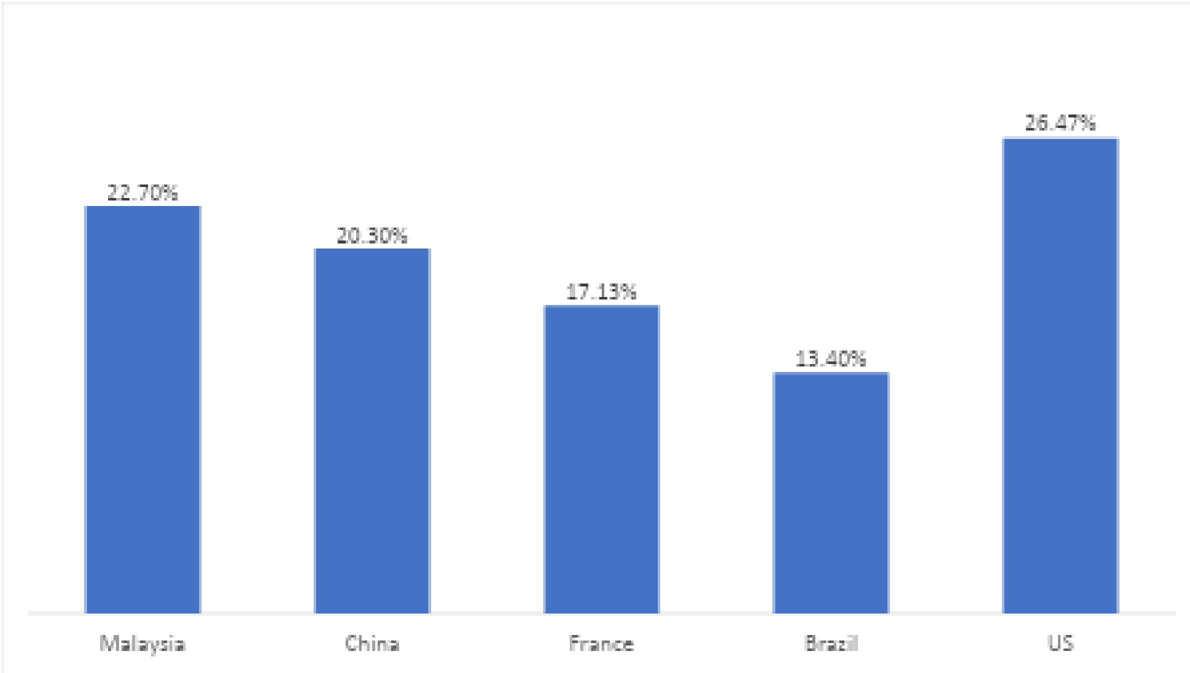


Figure 8.

Demand Percentage for Each Customers (United States included)



Proposal of Building a New Factory

The total forecast demand indicates that the total future demand is 2,669,450 units. With the current capacity of Austin and Paris of 600,000 each. They won't be able to handle the predicted demand from the forecast. Even though both factories expand the capacity to 1,200,000 units, the total capacity of 1,200,000 units is still lower than the forecast demand of 2,669,450 units. Therefore, it is recommended to build a new factory from the following candidate location: Australia, Charleston, India, Malaysia, Mobile, South African or Spain.

Linear Programming

With the information given by the Hanover Inc., constructing linear programming is chosen to look for the optimal solution for the business problem. The purpose of the optimization in production is to minimize the cost. The cost for producing the product for the customers include Fixed cost for operating the factory, variable cost for producing different products, shipping expenses for different customers, and duty cost for the selling price.

At the same time, the capacity from each factory is limited and can only be determined from two options: 60,000 units or 1,200,000 units. With all the variables, Hanover, Inc. needs to be certain that the expected production is equal or more than the forecast demand. While k is used to represent different products, i is used to represent different customers and j is used to represent different factories. For choosing the best candidate, F is also built in the program to represent whether the factory is active or not along with the fixed cost with specific capacity. The following formulas is the linear programming model:

$$\sum_{K} X_{kij} \sum F$$

The linear programming process is scripted by using python, the following section shows part of the output for the variable with indices looks like:

1245000.0*F_1 + 977000.0*F_2 + 1037000.0*F_3 + 1082000.0*F_4 + 960000.0*F_5 + 959000.0*F_6 + 1057000.0*F_7 + 801000.0*F_8 + 994000.0*F_9 + 4.5354*X_111 + 4.6454*X_112 + 3.855400*X_113 + 4.4454*X_114 + 3.7754*X_115 + 0.97*X_116 + 4.2554*X_117 + 4.105*X_118 + 4.6354*X_119 + 5.761*X_121 + 5.591*X_122 + 4.881*X_123 + 5.670*X_124 + 4.911*X_125 + 4.821*X_126 + 5.481*X_127 + 3.2555*X_128 + 5.701*X_129 + 3.0848*X_131 + 1.65*X_132 + 2.2948*X_133 + 2.9947*X_134 + 1.90136*X_135 + 1.99136*X_136 + 2.8048*X_137 + 1.92232*X_138 + 1.81*X_139 + 3.4024*X_141 + 3.31240*X_142 + 2.59239*X_143 + 3.3023*X_144 + ...

The model should also include the constraint:

1. The total quantity of production for a single factory should not exceed the capacity of 600, 000 or 1200,000.

2. The total quantity of production for products should be more or equal to the demand from the customer

Result and Recommendation

To support the analysis in determining whether Hanover, Inc should build a new factory for the upcoming demand, linear programming is applied to verify if the result is consistent with the proposal of building a new factory. The result indicates that it is infeasible to find an optimal solution, which indicates that the current factories cannot handle the forecast demand. The result of the linear programming supports our initial proposal of building a new factory(Figure 9).

Figure 9.

Infeasible output from Linear Programming

```
model.solve()
status=LpStatus[model.status]
[90]  ✓ 0.1s

status
[91]  ✓ 0.3s
... 'Infeasible'

print("Total Cost:", model.objective.value())

# Decision Variables

for v in model.variables():
    try:
        print(v.name,"=", v.value())
    except:
        print("error couldnt find value")
[92]  ✓ 0.4s
... Output exceeds the size limit. Open the full output data in a text editor
Total Cost: 10278914.880800001
F_1 = 1.0
F_2 = 1.0
F_3 = 0.0
```

From the previous analysis on products, customers and factories, the Hanover. Inc. appears to have higher BCR in expanding the existing factory instead of just maintaining the 600,000 units capacities. Increasing the capacity from 600,000 units to 1,200,000 may potentially help Hanover, Inc. to decrease the total cost. While the total demand from forecast is 2,669,450, the third factory only needs to have 600,000 to fulfil the demand. Therefore, there may be three different kinds of combination for the capacity of each factory:

1. Austin: 1,200,000; Paris: 1,200,000; candidate factory: 600,000

2. Austin: 1,200,000; Paris: 600,000; candidate factory: 1,200,000

3. Austin: 600,000; Paris: 1,200,000; candidate factory: 1,200,000

After inputting the different combination of capacities for the factories into the linear model, all the options can find an optimal solution. The result of the linear programming identifies the location of South Africa as the optimal choice that can fulfill the upcoming demand and minimize the cost.

The following table summarize the optimized solution with the criteria described above:
Option 1. Austin: 1,200,000; Paris: 1,200,000; South Africa: 600,000
The total cost: $9,307,337.052

			Austin	Paris	South Africa
Customer	Product	Demand	1,200,000	1,200,000	600,000
Malaysia	A	42900			42,900
	B	25800			25,800
	C	133160			133,160
China	A	318830			318,830
	B	57090			57,090
	C	165930			165,930
France	A	143600		143,600	

	B	31980		31,980	
	C	182080		182,080	
Brazil	A	19350			19,350
	B	27050			27,050
	C	77070			77,070
US Northeast	A	9870	9,870		
	B	34320	34,320		
	C	7830	7,830		
US Southeast	A	102970	102,970		
	B	71130	71,130		
	C	155230	155,230		
US Midwest	A	57490	57,490		
	B	175030	175,030		
	C	224750	224,750		
US West	A	183400	183,400		
	B	295230	295,230		
	C	127360	127,360		

Option 2. Austin: 1,200,000; Paris: 600,000; South Africa: 1,200,000

The total cost: $ 9,249,597.455

			Austin	Paris	South Africa
Customer	Product	Demand	1,200,000	600,000	1,200,000
Malaysia	A	42900			42,900
	B	25800			25800
	C	133160			133160
China	A	318830			318,830
	B	57090		57090	
	C	165930			165930
France	A	143600		143600	
	B	31980		31980	
	C	182080		182080	
Brazil	A	19350			19350
	B	27050			27050
	C	77070			77070
US Northeast	A	9870			9870
	B	34320	34320		
	C	7830	7830		

US Southeast	A	102970			102970
	B	71130	71130		
	C	155230	155230		
US Midwest	A	57490			57490
	B	175030	175030		
	C	224750	224750		
US West	A	183400	109,120		74,280
	B	295230	295230		
	C	127360	127360		

Option 3. Austin: 1,200,000; Paris: 600,000; South Africa: 1,200,000

The total cost: $ 9,208,486.151

			Austin	Paris	South Africa
Customer	Product	Demand	600000	1200,000	1200000
Malaysia	A	42900			42900
	B	25800			25800
	C	133160			133160
China	A	318830			318830
	B	57090			57090
	C	165930			165930
France	A	143600		143600	
	B	31980		31980	
	C	182080		182080	
Brazil	A	19350			19350
	B	27050			27050
	C	77070			77070
US Northeast	A	9870		9870	
	B	34320		34320	
	C	7830		7830	
US Southeast	A	102970		8660	94310
	B	71130		71130	
	C	155230		155230	
US Midwest	A	57490	2380		55110
	B	175030	175030		
	C	224750		224750	

US West	A	183400			183400
	B	295230	295230		
	C	127360	127360		

According to the total cost from three different options above, option 3 appears to be the cheapest amount of all the options. While the forecast demand remains the same as the production goal for all the options, the minimized cost from option 3 can assist Hanover Inc. to create the highest profit.

From the summary output for option 3, the Austin factory should produce 600,000 units, Paris should produce 869,450 units, and the South Africa factory should produce 1,200,000 units. It appears that both Austin and South Africa maximum their production capacity while the Paris factory does not. Although Paris has 1,200,000 units capacity for production, it will cost more for the Hanover, Inc to use the Paris factory to produce more products. With this optimization result, it is also recommended that South Africa produce for all the demand from Malaysia, China, and Brazil. While South Africa’s production cost for product A is approximately only 50% of the cost for Austin and Paris, it is also suggested for South Africa factory to produce most of the product A for United States

In conclusion, it is recommended for Hanover, Inc to build a new factory in South Africa with 1,200,000 units capacity to fulfill the forecast demand. However, Hanover, Inc. only provides limited information regarding building the new factory. The start-up expense for building a brand-new factory should cost the company more than the operation cost for the factory. Therefore, Hanover, Inc. should further investigate the possible expense associated with building a new factory and conduct another analysis with new information included in the future.

Appendix A

Linear Programming for Optimization

Model

725000.0*F_1 + 977000.0*F_2 + 1037000.0*F_3 + 1082000.0*F_4 + 960000.0*F_5 + 959000.0*F_6 + 1057000.0*F_7 + 1050000.0*F_8 + 994000.0*F_9 + 4.5354*X_111 + 4.6454*X_112 + 3.8554000000000004*X_113 + 4.4454*X_114 + 3.7754000000000003*X_115 + 0.9700000000000001*X_116 + 4.2554*X_117 + 4.1053999999999995*X_118 + 4.635400000000001*X_119 + 5.761*X_121 + 5.591*X_122 + 4.881*X_123 + 5.670999999999999*X_124 + 4.911*X_125 + 4.821*X_126 + 5.481*X_127 + 3.2555*X_128 + 5.7010000000000005*X_129 + 3.0848*X_131 + 1.6500000000000001*X_132 + 2.2948*X_133 + 2.9947999999999997*X_134 + 1.9013600000000002*X_135 + 1.99136*X_136 + 2.8048*X_137 + 1.92232*X_138 + 1.81*X_139 + 3.4024*X_141 + 3.3124000000000002*X_142 + 2.5923999999999996*X_143 + 3.3023999999999996*X_144 + 2.7724*X_145 + 2.8423999999999996*X_146 + 3.1124*X_147 + 2.3924*X_148 + 3.3324*X_149 + 1.73*X_151 + 2.3674600000000003*X_152 + 2.0174600000000003*X_153 + 1.63*X_154 + 1.76746*X_155 + 1.65746*X_156 + 1.44*X_157 + 2.0174600000000003*X_158 + 2.50746*X_159 + 1.67*X_161 + 2.3674600000000003*X_162 + 2.0174600000000003*X_163 + 1.5699999999999998*X_164 + 1.76746*X_165 + 1.6674600000000002*X_166 + 1.38*X_167 + 2.0174600000000003*X_168 + 2.50746*X_169 + 1.65*X_171 + 2.39746*X_172 + 2.03746*X_173 + 1.55*X_174 + 1.7874599999999998*X_175 + 1.67746*X_176 + 1.36*X_177 + 2.02746*X_178 + 2.52746*X_179 + 1.66*X_181 + 2.50746*X_182 + 2.02746*X_183 + 1.56*X_184 + 1.76746*X_185 + 1.6674600000000002*X_186 + 1.37*X_187 + 2.0374600000000003*X_188 + 2.5374600000000003*X_189 + 4.4826*X_211 + 4.602600000000001*X_212 + 3.9626*X_213 + 4.6926000000000005*X_214 + 3.8726000000000003*X_215 + 1.13*X_216 + 4.2026*X_217 + 4.2126*X_218 + 4.9026*X_219 + 5.689*X_221 + 5.529*X_222 + 4.968999999999999*X_223 + 5.899*X_224 + 4.989*X_225 + 4.909*X_226 + 5.409*X_227 + 3.3795*X_228 + 5.949*X_229 + 3.0604999999999998*X_231 + 1.6600000000000001*X_232 + 2.4305*X_233 + 3.2704999999999997*X_234 + 2.03435*X_235 + 2.13435*X_236 + 2.7805*X_237 + 2.0672*X_238 + 2.13*X_239 + 3.3735999999999997*X_241 + 3.2935999999999996*X_242 + 2.7236000000000002*X_243 + 3.5736*X_244 + 2.8936*X_245 + 2.9736000000000002*X_246 + 3.0835999999999997*X_247 + 2.5236*X_248 + 3.6235999999999997*X_249 + 1.73*X_251 + 2.36864*X_252 + 2.16864*X_253 + 1.93*X_254 + 1.9086400000000001*X_255 + 1.80864*X_256 + 1.44*X_257 + 2.16864*X_258 + 2.81864*X_259 + 1.67*X_261 + 2.36864*X_262 + 2.16864*X_263 + 1.87*X_264 + 1.9086400000000001*X_265 + 1.8186399999999998*X_266 + 1.38*X_267 + 2.16864*X_268 + 2.81864*X_269 + 1.65*X_271 + 2.39864*X_272 + 2.18864*X_273 + 1.85*X_274 + 1.9286400000000001*X_275 + 1.82864*X_276 + 1.36*X_277 + 2.17864*X_278 + 2.83864*X_279 + 1.66*X_281 + 2.5086399999999998*X_282 + 2.17864*X_283 + 1.8599999999999999*X_284 + 1.9086400000000001*X_285 + 1.8186399999999998*X_286 + 1.37*X_287 + 2.18864*X_288

+ 2.8486399999999996*X_289 + 4.6548*X_311 + 4.694800000000001*X_312 + 4.0548*X_313 + 4.7648*X_314 + 3.9648000000000003*X_315 + 1.1199999999999999*X_316 + 4.7148*X_317 + 4.3048*X_318 + 4.9948*X_319 + 5.901999999999999*X_321 + 5.662*X_322 + 5.102*X_323 + 6.0120000000000005*X_324 + 5.122*X_325 + 5.052*X_326 + 5.962*X_327 + 3.436*X_328 + 6.082*X_329 + 3.1664000000000003*X_331 + 1.6400000000000001*X_332 + 2.4564000000000004*X_333 + 3.2763999999999998*X_334 + 2.04648*X_335 + 2.15648*X_336 + 3.2264*X_337 + 2.0757600000000003*X_338 + 2.1100000000000003*X_339 + 3.4947999999999997*X_341 + 3.3347999999999995*X_342 + 2.7648*X_343 + 3.5947999999999993*X_344 + 2.9348*X_345 + 3.0248*X_346 + 3.5447999999999995*X_347 + 2.5648*X_348 + 3.6647999999999996*X_349 + 1.79*X_351 + 2.3688000000000002*X_352 + 2.1688*X_353 + 1.89*X_354 + 1.9088000000000003*X_355 + 1.8188*X_356 + 1.8399999999999999*X_357 + 2.1688*X_358 + 2.8188*X_359 + 1.73*X_361 + 2.3688000000000002*X_362 + 2.1688*X_363 + 1.83*X_364 + 1.9088000000000003*X_365 + 1.8288000000000002*X_366 + 1.7799999999999998*X_367 + 2.1688*X_368 + 2.8188*X_369 + 1.71*X_371 + 2.3988*X_372 + 2.1888*X_373 + 1.81*X_374 + 1.9288*X_375 + 1.8388*X_376 + 1.76*X_377 + 2.1788000000000003*X_378 + 2.8388000000000004*X_379 + 1.72*X_381 + 2.5088000000000004*X_382 + 2.1788000000000003*X_383 + 1.8199999999999998*X_384 + 1.9088000000000003*X_385 + 1.8288000000000002*X_386 + 1.77*X_387 + 2.1888*X_388 + 2.8488*X_389 + 0.0

Capacity_constrains0:
X_111 + X_121 + X_131 + X_141 + X_151 + X_161 + X_171
+ X_181 + X_211 + X_221 + X_231 + X_241 + X_251 + X_261 + X_271 + X_281
+ X_311 + X_321 + X_331 + X_341 + X_351 + X_361 + X_371 + X_381 <= 600000

Capacity_constrains1: X_112 + X_122 + X_132 + X_142 + X_152 + X_162 + X_172
+ X_182 + X_212 + X_222 + X_232 + X_242 + X_252 + X_262 + X_272 + X_282
+ X_312 + X_322 + X_332 + X_342 + X_352 + X_362 + X_372 + X_382 <= 1200000

Capacity_constrains2: X_113 + X_123 + X_133 + X_143 + X_153 + X_163 + X_173
+ X_183 + X_213 + X_223 + X_233 + X_243 + X_253 + X_263 + X_273 + X_283
+ X_313 + X_323 + X_333 + X_343 + X_353 + X_363 + X_373 + X_383 <= 600000

Capacity_constrains3: X_114 + X_124 + X_134 + X_144 + X_154 + X_164 + X_174
+ X_184 + X_214 + X_224 + X_234 + X_244 + X_254 + X_264 + X_274 + X_284
+ X_314 + X_324 + X_334 + X_344 + X_354 + X_364 + X_374 + X_384 <= 600000

Capacity_constrains4: X_115 + X_125 + X_135 + X_145 + X_155 + X_165 + X_175
+ X_185 + X_215 + X_225 + X_235 + X_245 + X_255 + X_265 + X_275 + X_285
+ X_315 + X_325 + X_335 + X_345 + X_355 + X_365 + X_375 + X_385 <= 600000

Capacity_constrains5: X_116 + X_126 + X_136 + X_146 + X_156 + X_166 + X_176
+ X_186 + X_216 + X_226 + X_236 + X_246 + X_256 + X_266 + X_276 + X_286

+ X_316 + X_326 + X_336 + X_346 + X_356 + X_366 + X_376 + X_386 <= 600000

Capacity_constrains6: X_117 + X_127 + X_137 + X_147 + X_157 + X_167 + X_177 + X_187 + X_217 + X_227 + X_237 + X_247 + X_257 + X_267 + X_277 + X_287 + X_317 + X_327 + X_337 + X_347 + X_357 + X_367 + X_377 + X_387 <= 600000

Capacity_constrains7: X_118 + X_128 + X_138 + X_148 + X_158 + X_168 + X_178 + X_188 + X_218 + X_228 + X_238 + X_248 + X_258 + X_268 + X_278 + X_288 + X_318 + X_328 + X_338 + X_348 + X_358 + X_368 + X_378 + X_388 <= 1200000

Capacity_constrains8: X_119 + X_129 + X_139 + X_149 + X_159 + X_169 + X_179 + X_189 + X_219 + X_229 + X_239 + X_249 + X_259 + X_269 + X_279 + X_289 + X_319 + X_329 + X_339 + X_349 + X_359 + X_369 + X_379 + X_389 <= 600000

Demand_Constraints00: X_111 + X_112 + X_113 + X_114 + X_115 + X_116 + X_117 + X_118 + X_119 >= 42900

Demand_Constraints10: X_121 + X_122 + X_123 + X_124 + X_125 + X_126 + X_127 + X_128 + X_129 >= 318830

Demand_Constraints20: X_131 + X_132 + X_133 + X_134 + X_135 + X_136 + X_137 + X_138 + X_139 >= 143600

Demand_Constraints30: X_141 + X_142 + X_143 + X_144 + X_145 + X_146 + X_147 + X_148 + X_149 >= 19350

Demand_Constraints40: X_151 + X_152 + X_153 + X_154 + X_155 + X_156 + X_157 + X_158 + X_159 >= 9870

Demand_Constraints50: X_161 + X_162 + X_163 + X_164 + X_165 + X_166 + X_167 + X_168 + X_169 >= 102970

Demand_Constraints60: X_171 + X_172 + X_173 + X_174 + X_175 + X_176 + X_177 + X_178 + X_179 >= 57490

Demand_Constraints70: X_181 + X_182 + X_183 + X_184 + X_185 + X_186 + X_187 + X_188 + X_189 >= 183400

Demand_Constraints01: X_211 + X_212 + X_213 + X_214 + X_215 + X_216 + X_217 + X_218 + X_219 >= 25800

Demand_Constraints11: X_221 + X_222 + X_223 + X_224 + X_225 + X_226 + X_227 + X_228 + X_229 >= 57090

Demand_Constraints21: X_231 + X_232 + X_233 + X_234 + X_235 + X_236 + X_237 + X_238 + X_239 >= 31980

Demand_Constraints31: X_241 + X_242 + X_243 + X_244 + X_245 + X_246 + X_247 + X_248 + X_249 >= 27050

Demand_Constraints41: X_251 + X_252 + X_253 + X_254 + X_255 + X_256 + X_257 + X_258 + X_259 >= 34320

Demand_Constraints51: X_261 + X_262 + X_263 + X_264 + X_265 + X_266 + X_267 + X_268 + X_269 >= 71130

Demand_Constraints61: X_271 + X_272 + X_273 + X_274 + X_275 + X_276 + X_277 + X_278 + X_279 >= 175030

Demand_Constraints71: X_281 + X_282 + X_283 + X_284 + X_285 + X_286 + X_287 + X_288 + X_289 >= 295230

Demand_Constraints02: X_311 + X_312 + X_313 + X_314 + X_315 + X_316 + X_317 + X_318 + X_319 >= 133160

Demand_Constraints12: X_321 + X_322 + X_323 + X_324 + X_325 + X_326 + X_327 + X_328 + X_329 >= 165930

Demand_Constraints22: X_331 + X_332 + X_333 + X_334 + X_335 + X_336 + X_337 + X_338 + X_339 >= 182080

Demand_Constraints32: X_341 + X_342 + X_343 + X_344 + X_345 + X_346 + X_347 + X_348 + X_349 >= 77070

Demand_Constraints42: X_351 + X_352 + X_353 + X_354 + X_355 + X_356 + X_357 + X_358 + X_359 >= 7830

Demand_Constraints52: X_361 + X_362 + X_363 + X_364 + X_365 + X_366 + X_367 + X_368 + X_369 >= 155230

Demand_Constraints62: X_371 + X_372 + X_373 + X_374 + X_375 + X_376 + X_377 + X_378 + X_379 >= 224750

Demand_Constraints72: X_381 + X_382 + X_383 + X_384 + X_385 + X_386 + X_387 + X_388 + X_389 >= 127360

factory1_active_constrains: F_1 = 1

factory2_active_constrains: F_2 = 1

3_factory_active_constraints: F_1 + F_2 + F_3 + F_4 + F_5 + F_6 + F_7 + F_8 + F_9 = 3

_C1: - 1e+15 F_1 + X_111 + X_121 + X_131 + X_141 + X_151 + X_161 + X_171
+ X_181 + X_211 + X_221 + X_231 + X_241 + X_251 + X_261 + X_271 + X_281
+ X_311 + X_321 + X_331 + X_341 + X_351 + X_361 + X_371 + X_381 <= 0

_C2: - 1e+15 F_2 + X_112 + X_122 + X_132 + X_142 + X_152 + X_162 + X_172
+ X_182 + X_212 + X_222 + X_232 + X_242 + X_252 + X_262 + X_272 + X_282
+ X_312 + X_322 + X_332 + X_342 + X_352 + X_362 + X_372 + X_382 <= 0

_C3: - 1e+15 F_3 + X_113 + X_123 + X_133 + X_143 + X_153 + X_163 + X_173
+ X_183 + X_213 + X_223 + X_233 + X_243 + X_253 + X_263 + X_273 + X_283
+ X_313 + X_323 + X_333 + X_343 + X_353 + X_363 + X_373 + X_383 <= 0

_C4: - 1e+15 F_4 + X_114 + X_124 + X_134 + X_144 + X_154 + X_164 + X_174
+ X_184 + X_214 + X_224 + X_234 + X_244 + X_254 + X_264 + X_274 + X_284
+ X_314 + X_324 + X_334 + X_344 + X_354 + X_364 + X_374 + X_384 <= 0

_C5: - 1e+15 F_5 + X_115 + X_125 + X_135 + X_145 + X_155 + X_165 + X_175
+ X_185 + X_215 + X_225 + X_235 + X_245 + X_255 + X_265 + X_275 + X_285
+ X_315 + X_325 + X_335 + X_345 + X_355 + X_365 + X_375 + X_385 <= 0

_C6: - 1e+15 F_6 + X_116 + X_126 + X_136 + X_146 + X_156 + X_166 + X_176
+ X_186 + X_216 + X_226 + X_236 + X_246 + X_256 + X_266 + X_276 + X_286
+ X_316 + X_326 + X_336 + X_346 + X_356 + X_366 + X_376 + X_386 <= 0

_C7: - 1e+15 F_7 + X_117 + X_127 + X_137 + X_147 + X_157 + X_167 + X_177
+ X_187 + X_217 + X_227 + X_237 + X_247 + X_257 + X_267 + X_277 + X_287
+ X_317 + X_327 + X_337 + X_347 + X_357 + X_367 + X_377 + X_387 <= 0

_C8: - 1e+15 F_8 + X_118 + X_128 + X_138 + X_148 + X_158 + X_168 + X_178
+ X_188 + X_218 + X_228 + X_238 + X_248 + X_258 + X_268 + X_278 + X_288
+ X_318 + X_328 + X_338 + X_348 + X_358 + X_368 + X_378 + X_388 <= 0

_C9: - 1e+15 F_9 + X_119 + X_129 + X_139 + X_149 + X_159 + X_169 + X_179
+ X_189 + X_219 + X_229 + X_239 + X_249 + X_259 + X_269 + X_279 + X_289
+ X_319 + X_329 + X_339 + X_349 + X_359 + X_369 + X_379 + X_389 <= 0

VARIABLES
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